

Retail Customer Analysis

Phase 2 — Data Cleaning & Database Preparation

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Overview

With the data understood, this phase prepared it for analysis. Working from the issues identified in Phase 1, we corrected and organised the records through a careful, step-by-step process — removing duplicates, setting aside returns and non-product charges, and discarding invalid entries. The count of records was tracked after every step, so the effect of each decision is fully transparent. The cleaned data was then organised into structured tables and loaded into a database.

Results at a Glance

94% RECORDS RETAINED	1.00M CLEAN SALES RECORDS	5,852 IDENTIFIED CUSTOMERS	19,104 RETURNS SET ASIDE	3 DATABASE TABLES
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Key Findings

1. The data was high quality — 94% of records were kept.

Of 1,067,371 original records, 1,003,214 survived cleaning. Only about 6% were removed, confirming a reliable dataset that needs little correction.

2. Duplicate records were the single biggest issue.

34,335 exact duplicate records were removed first. Left in place, they would have inflated every sales figure.

3. Returns were preserved, not discarded.

19,104 cancellation records were moved into a separate table so they can be analysed in their own right, rather than being deleted or mixed in with genuine sales.

4. Non-product charges were filtered out.

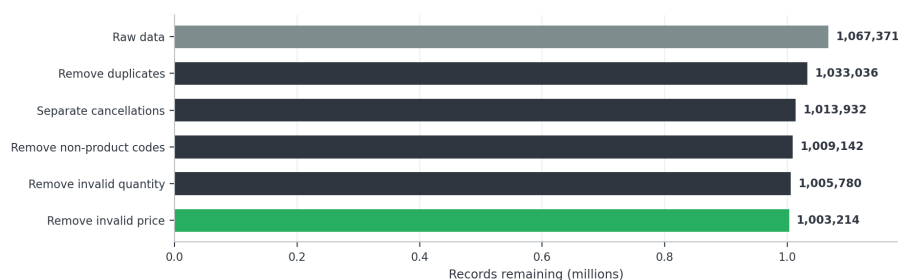
4,790 records for postage, manual adjustments, fees and bad-debt entries were set aside so that product analysis reflects actual goods sold.

5. Customer analysis will cover 5,852 customers.

776,577 records carry a customer identifier and form the basis for loyalty, value and retention analysis; the remainder (guest checkouts) are retained for product and revenue analysis.

How the Data Was Cleaned

Records remaining after each step. Small reductions at each stage show cleaning was targeted, not aggressive.



What's Next — Phase 3

Phase 3 begins the analysis with a cohort study: grouping customers by the month they first purchased and tracking how many keep buying over time. This reveals how well the business retains its customers — one of the clearest indicators of long-term health.